ECO 319/619
Time Series Analysis and Forecasting Presentation

Topic - Effects of EV (Electric Vehicle) sales on Gasoline Prices in India

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Motivation:

- There is a fast transition from fossil fuel based vehicles to electric vehicles in India and this is a major step towards greener mobility revolution and will help our country to achieve decarbonization goals
- Plug-in hybrid electric vehicles (PHEVs), extended-range battery electric vehicles (E-REVs), battery electric vehicles (BEVs), and hybrid electric vehicles (HEVs) are collectively referred to as EVs. They rely on electricity instead of fossil fuel thus reducing greenhouse gas emission
- Due to uncertainty in the oil pricing and availability of crude oil, countries from all over the world are taking steps towards adoption of EVs by introducing various policies.
- India on the similar lines has set the target of adopting 30% of its vehicles in the electric segment by 2030 and 100% by 2047
- While adoption of EVs comes with certain pros and cons and face of the barrier of people's attitude and bias, there is consistent increase in the sales of EVs in India.
- This increase in sales of EVs can impact gasoline prices. This research helps to study the impact of EV sales on gasoline prices

Novelty of the Research:

1. This study tries to analyze impact of recently emerging EV industry on traditional gasoline prices

2. The data used for above study is recent data and helps us in understanding the current trends in EV industry

3. The study is exclusively carried out in India and analyzes EV sales and gasoline prices both from an Indian perspective.

4. The study also takes into account the effect of Russia Ukraine war on Indian gasoline prices and its effects on sales of EV in India.

Data:

The data span ranges from January 2015 to June 2025. The variables used for the above study includes:

1. Electric Vehicles sales in India (Plug-in hybrid EV, Electric(Battery Operated Vehicles) , Pure EV, Strong hybrid EV)
   Source: Vahan Sewa
2. Gasoline Prices (INR)
   Source: Indexmundi
3. Exchange rate (LAV in %)
   Source: FEDAI (Foreign Exchange Dealers' Association of India)
4. CPI
   Source: MOSPI
5. Brent Crude Oil Price (INR)
   Source: Macrotrends

The data covers important events and helps us in studying the effects of FAME 1 and FAME 2 policies, COVID and effect of Russia Ukraine war. As India imports significant portion of its oil imports from Russia (36%) , the Russia Ukraine war can impact oil prices in India.

Methodology:

- We will be using R Studio for analysis and forecasting.
- As in real world economy one variable affects every other variable, each variable shares cause effect relationship which can be captured by VAR model. As residuals are not independent over time, there exists a serial correlation in data. Also skewness show significant asymmetry and kurtosis show extreme values (heavy tails). Therefore we use Bayesian VAR which treats model parameters as random with probability distribution instead of normal VAR which treats parameters as fixed values.

Dummies used:

- Dummy for FAME 1 ((Faster Adoption and Manufacturing of Hybrid and Electric Vehicles): Launched by Indian government from April 1, 2015 to March 31, 2019
- Dummy for FAME 2 ((Faster Adoption and Manufacturing of Hybrid and Electric Vehicles): Launched by Indian government from April 1, 2019 to March 31, 2024
- Dummy for Covid effect: From March 2020
- Dummy for Russia Ukraine war: Russia's Ukraine invasion began in February 2014 and got converted to full scale war from February, 2022

These dummies will be considered as external shock to the system

Methodology:

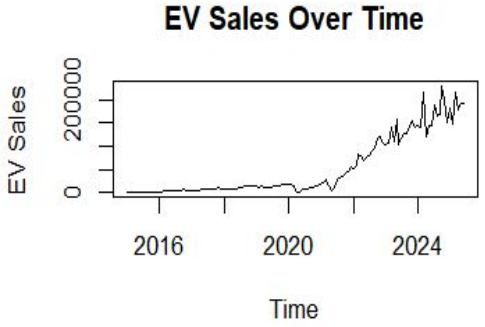
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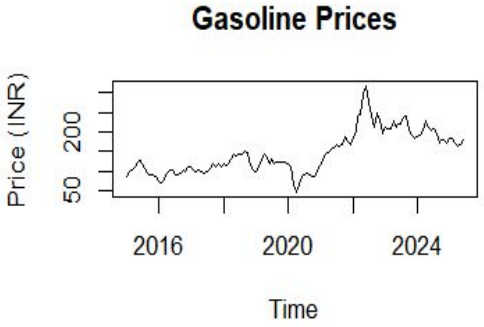
```
> summary(ev_data)
     YEAR              TOTAL_EV       GASOLINE PRICES (INR) EXCHANGE RATE (LAV IN %)
 Length:126         Min.   :   168   Min.   : 45.18         Min.   :12.14
 Class :character   1st Qu.:  6631   1st Qu.:102.64         1st Qu.:12.47
 Mode  :character   Median : 14089   Median :128.09         Median :12.49
                    Mean   : 52123   Mean   :146.68         Mean   :12.45
                    3rd Qu.:102830   3rd Qu.:185.64         3rd Qu.:12.49
                    Max.   :229673   Max.   :318.72         Max.   :12.49
 BRENT CRUDE OIL PRICE(INR)      CPI
 Min.   : 1318              Min.   :119.5
 1st Qu.: 4618              1st Qu.:134.0
 Median : 5924              Median :150.7
 Mean   : 5909              Mean   :154.6
 3rd Qu.: 6903              3rd Qu.:175.7
 Max.   :11140              Max.   :196.5
```

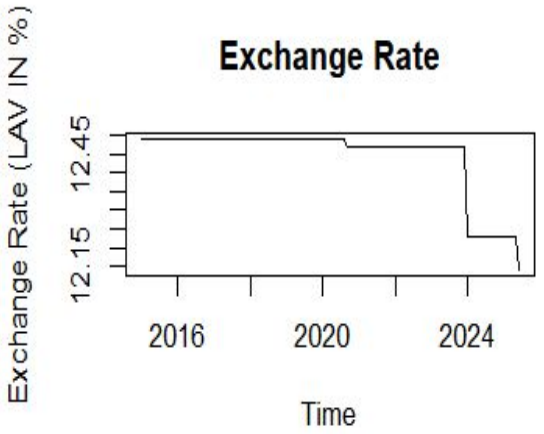
```
> summary(dummies)
     FAME1            FAME2            COVID          RUS_UKR_WAR
 Min.   :0.000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
 1st Qu.:0.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
 Median :0.000   Median :0.0000   Median :1.0000   Median :0.0000
 Mean   :0.381   Mean   :0.4762   Mean   :0.5079   Mean   :0.3254
 3rd Qu.:1.000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000
 Max.   :1.000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000
```

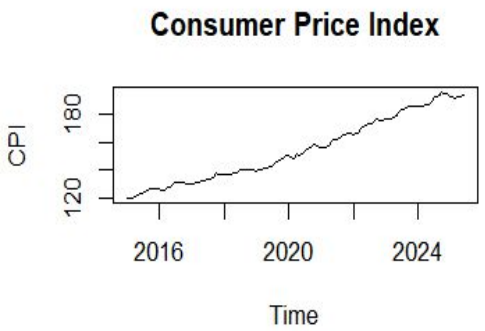
Methodology:

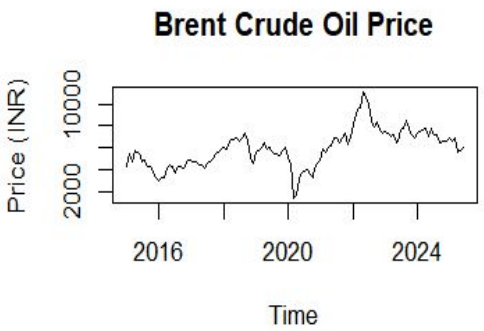
Plotting the time series data











Methodology:

The results of correlation matrix are as follows:

- Correlation between TOTAL_EV and CPI is +0.907 indicating that EV sales are highly linked to overall economic condition.
- Correlation between TOTAL_EV and EXCHANGE RATE IS -0.806 stating a higher EV adoption could be associated with weaker currency or specific policy
- Correlation between TOTAL_EV and GASOLINE PRICES is +0.712, indicating higher fuel costs boost EV sales
- Correlation between TOTAL_EV and BRENT CRUDE OIL PRICE is +0.554, indicating moderate relationship. Thus gasoline prices affect EV sales directly.

	TOTAL_EV	GASOLINE PRICES (INR)	EXCHANGE RATE (LAV IN %)
TOTAL_EV	1.0000000	0.7116028	-0.8063883
GASOLINE PRICES (INR)	0.7116028	1.0000000	-0.3802560
EXCHANGE RATE (LAV IN %)	-0.8063883	-0.3802560	1.0000000
BRENT CRUDE OIL PRICE(INR)	0.5537372	0.9290346	-0.2776099
CPI	0.9072351	0.7631661	-0.7066925

	BRENT CRUDE OIL PRICE(INR)	CPI
TOTAL_EV	0.5537372	0.9072351
GASOLINE PRICES (INR)	0.9290346	0.7631661
EXCHANGE RATE (LAV IN %)	-0.2776099	-0.7066925
BRENT CRUDE OIL PRICE(INR)	1.0000000	0.6107497
CPI	0.6107497	1.0000000

Methodology:

Checking for stationarity using ADF test:

- Null Hypothesis (H_0): The time series has a unit root (i.e., it is non-stationary)
- Alternative Hypothesis (H_1): The time series is stationary
- Decision Rule: If p-value > 0.05, fail to reject H_0 (non-stationary)

The results of ADF test are as follows:

- TOTAL_EV: p-value = 0.9578 → Non-stationary
- GASOLINE PRICES: p-value = 0.4442 → Non-stationary
- EXCHANGE RATE: p-value = 0.9781 → Non-stationary
- BRENT CRUDE OIL: p-value = 0.3925 → Non-stationary
- CPI: p-value = 0.5797 → Non-stationary

As all variables are non stationary and we will need differencing.
After doing first differencing, we obtain p-value of all variables = 0.01.
Thus the data is stationary after first differencing

Results:

The results from Bayesian VAR model fit are as follows:

- The model analyzes how Total EV sales, Gasoline Prices, the Exchange Rate, Brent Crude Oil Prices, and the CPI influence each other over time.
- The model consists of 121 observations, 5 variables each with 4 lags, hyperparameter sets lambda at 0.009 automatically and posterior marginal likelihood is given by -2928.548
- Log likelihood of the model is -2707.359 while MCMC draws =10,000

Results:

```
> summary(bvar_with_dummies)
Bayesian VAR consisting of 121 observations, 5 variables and 4 lags.
Time spent calculating: 2.46 secs
Hyperparameters: lambda
Hyperparameter values after optimisation: 0.009
Iterations (burnt / thinning): 10000 (5000 / 1)
Accepted draws (rate): 771 (0.154)

Numeric array (dimensions 21, 5) of coefficient values from a BVAR.
Median values:
                                TOTAL_EV GASOLINE PRICES (INR) EXCHANGE RATE (LAV IN %)
constant                        3978.848                -0.175                       0
TOTAL_EV-lag1                     -0.735                 0.000                       0
GASOLINE PRICES (INR)-lag1        -8.187                 0.916                       0
EXCHANGE RATE (LAV IN %)-lag1     14.868                -0.002                       1
BRENT CRUDE OIL PRICE(INR)-lag1    0.823                -0.002                       0
CPI-lag1                          32.274                 0.008                       0
TOTAL_EV-lag2                     -0.412                 0.000                       0
GASOLINE PRICES (INR)-lag2        -3.894                -0.006                       0
EXCHANGE RATE (LAV IN %)-lag2     -8.010                 0.003                       0
BRENT CRUDE OIL PRICE(INR)-lag2    0.772                -0.006                       0
CPI-lag2                          -3.519                -0.008                       0
TOTAL_EV-lag3                     -0.310                 0.000                       0
GASOLINE PRICES (INR)-lag3        -1.195                 0.001                       0
EXCHANGE RATE (LAV IN %)-lag3      4.076                -0.005                       0
BRENT CRUDE OIL PRICE(INR)-lag3   -0.191                 0.000                       0
CPI-lag3                           0.808                -0.003                       0
TOTAL_EV-lag4                     -0.082                 0.000                       0
GASOLINE PRICES (INR)-lag4         0.568                 0.000                       0
EXCHANGE RATE (LAV IN %)-lag4     -3.179                 0.004                       0
BRENT CRUDE OIL PRICE(INR)-lag4    0.263                 0.000                       0
```

Results:

	BRENT CRUDE OIL PRICE(INR)	CPI
constant	1.398	0.006
TOTAL_EV-lag1	-0.002	0.000
GASOLINE PRICES (INR)-lag1	-0.661	0.000
EXCHANGE RATE (LAV IN %)-lag1	-1.008	0.000
BRENT CRUDE OIL PRICE(INR)-lag1	0.264	0.000
CPI-lag1	0.834	0.992
TOTAL_EV-lag2	-0.002	0.000
GASOLINE PRICES (INR)-lag2	-0.078	0.000
EXCHANGE RATE (LAV IN %)-lag2	0.152	0.000
BRENT CRUDE OIL PRICE(INR)-lag2	-0.051	0.000
CPI-lag2	-0.172	0.000
TOTAL_EV-lag3	0.002	0.000
GASOLINE PRICES (INR)-lag3	-0.036	0.000
EXCHANGE RATE (LAV IN %)-lag3	-0.435	0.000
BRENT CRUDE OIL PRICE(INR)-lag3	-0.044	0.000
CPI-lag3	-0.055	0.000
TOTAL_EV-lag4	0.002	0.000
GASOLINE PRICES (INR)-lag4	-0.013	0.000
EXCHANGE RATE (LAV IN %)-lag4	0.078	0.000
BRENT CRUDE OIL PRICE(INR)-lag4	-0.016	0.000
CPI-lag4	-0.017	0.000

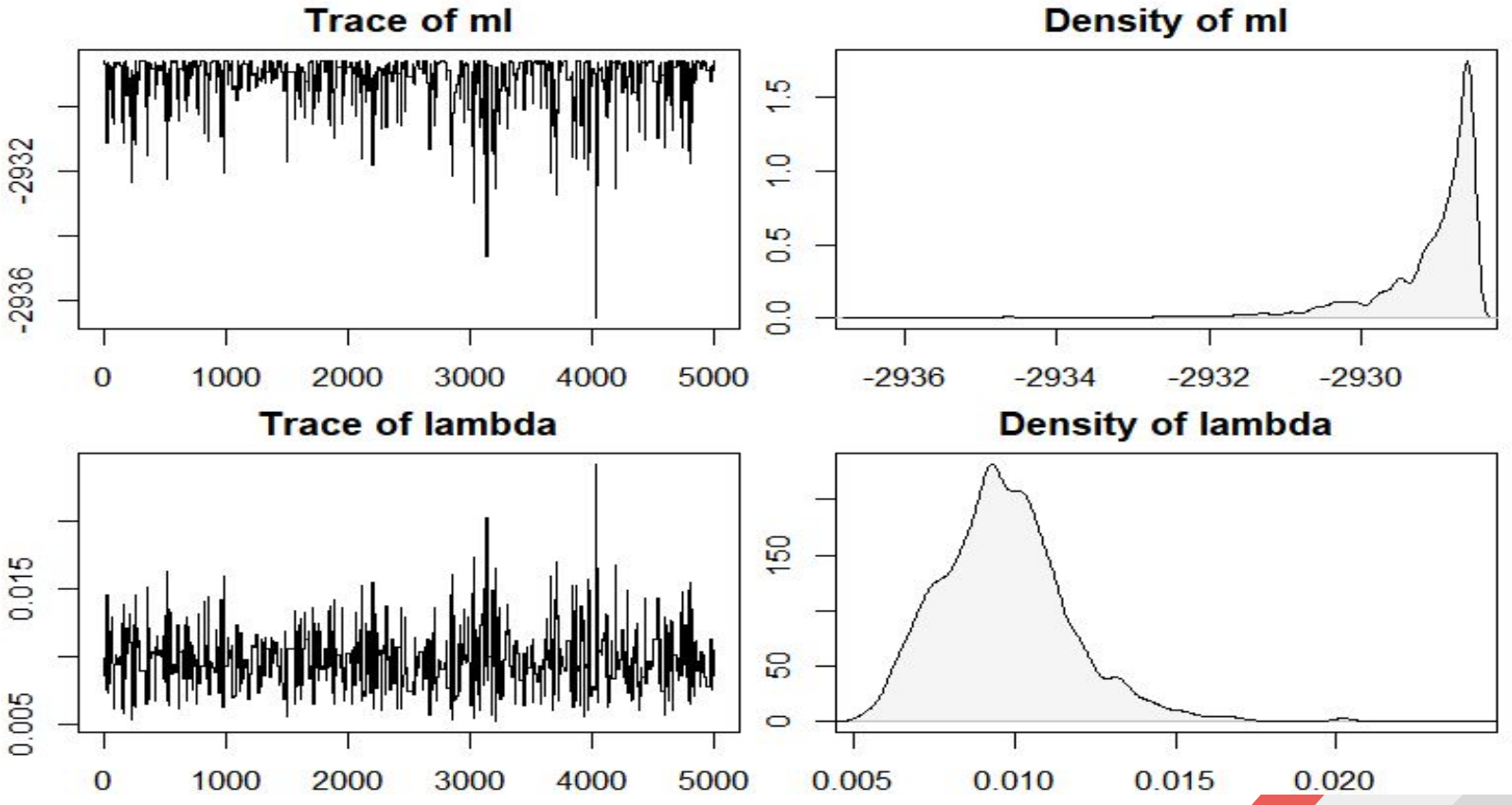
Numeric array (dimensions 5, 5) of variance-covariance values from a BVAR.
Median values:

	TOTAL_EV	GASOLINE PRICES (INR)	EXCHANGE RATE (LAV IN %)
TOTAL_EV	220474234.531	10112.664	-16.438
GASOLINE PRICES (INR)	10112.664	286.512	0.001
EXCHANGE RATE (LAV IN %)	-16.438	0.001	0.001
BRENT CRUDE OIL PRICE(INR)	-769865.673	5602.155	-1.227
CPI	-1901.965	-0.225	0.000

	BRENT CRUDE OIL PRICE(INR)	CPI
TOTAL_EV	-769865.673	-1901.965
GASOLINE PRICES (INR)	5602.155	-0.225
EXCHANGE RATE (LAV IN %)	-1.227	0.000
BRENT CRUDE OIL PRICE(INR)	391500.224	-22.430
CPI	-22.430	1.274

Log-Likelihood: -2707.359

Robustness Check:



Interpretation:

TOTAL_EV:

- The lags initially show negative values then slowly shifting towards positive values in its own equation indicating mean reversion leading to mild slowdown in sales after past spikes in EV sales.
- All zero values of lags in gasoline price equation state EV sales do not impact gasoline prices in short run. Same trend is observed in case of CPI
- Similar to gasoline prices, EV sales do not influence exchange rate leading to zero lag values
- The small negative values of lags in brent crude oil price equation which later on turn positive gives us weak evidence that rising EV sales might slightly reduce oil prices in future.

GASOLINE PRICES (INR)

- The first three values of lags in TOTAL_EV states that as gasoline prices rise EV sales falls, while fourth positive lag states that EV sales recover after some time. This might suggest that consumers take time for adjustment leading to short run inverse relationship and later leading to correction where consumers shift to EV considering high fuel prices
- A very high positive first lag in its own equation shows that gasoline prices are sticky and highly reflect its own past month prices. Prices beyond one month have little to no impact

Interpretation:

- Exchange rate and CPI are not influenced much by changes in gasoline prices leading to almost zero lags
- Gasoline prices show all negative lags in Brent crude oil price equation. This is likely due to the fact that when crude oil prices fall, domestic prices adjust to it with delay, creating negative link

EXCHANGE RATE(LAV IN %)

- The first lag in TOTAL_EV is highly positive i.e. when rupee depreciates in previous month, EV sales rises in current month likely due to the fact weaker rupee raisis import cost for oil leading to increase in fuel prices leading consumers to shift towards EVs. Second negative lags shows increase in EV sales due to rupee depreciation is not sustained. Third and fourth lags with alternate signs show short term responses to macro shocks.
- The tiny values of lags with alternating sign in gasoline price prices show that short run monthly fluctuations in INR/USD don't significantly affect domestic fuel prices much.
- In its own equation, first lag is 1 and all other lags are zero showing INR/USD rate moves slowly and today's value is highly affected by last month's value.
- The zero values of lags in CPI shows short run currency fluctuations don't change inflation immediately.

Interpretation:

- The alternating negative and positive values of lags in brent crude oil equation with weak magnitude shows weak rupee at lag 1 correlates with lower crude prices mostly due to reverse causality i.e. crude price drops tend to strengthen rupee later. Thus there seems no consistent or causal relationship in short run.

BRENT CRUDE OIL PRICE (INR)

- Initial positive, slight negative and again positive values of lags in TOTAL_EV shows EV sales rises when oil prices rise, some short term correction and some continuing effects
- The values of lags in GASOLINE PRICE equation almost tend to zero suggesting gasoline prices are weakly and slowly linked to global crude oil prices
- The lags of BRENT CRUDE OIL PRICE has little to no impact on exchange rate
- Strong positive first lag in its own equation while other lags are negative, indicating short run momentum followed by some mean reversion
- No impact on CPI in short run leading to zero lag values

CPI

- It has mixed and economically small effect on sales of EV. Higer first lag suggests that high inflation month ago might increase EV sales today but no clear evidence

Interpretation:

- Positive first lag in GASOLINE PRICE suggest high inflation last month could increase gasoline price this month, other lag values are negative and small suggesting mean reversion
- Inflation does not show any short term linear relationship with exchange rate, hence all lag values are zero in EXCHANGE RATE.
- First lag is positive in BRENT CRUDE OIL PRICE while all other lags are negative suggesting complex relationship with unclear transmission channel from domestic inflation to global oil prices.
- In its own equation, first lag is strong and positive while all other lags are zero indicating the primary driver of short-term inflation in the model is its own past value.

The results from variance covariance matrix are as follows:

- CRUDE: 391,500.224 indicating high variance
- TOTAL_EV: 220,474,234.531 indicating extremely high variance
- GASOLINE PRICES: 286.512 indicating moderate variance
- EXCHANGE RATE: 0.001 indicating very low variance
- CPI: 1.274 indicating very low variance

TOTAL_EV and BRENT CRUDE OIL PRICE has highest residual volatility

Interpretation:

TOTAL_EV with GASOLINE PRICE shows a positive covariance of 10,112.664 indicating that unexpected increases in EV sales tend to occur with unexpected increases in gasoline prices. This might be due to the fact that boost in economic activity is driving both up.

Results from dummy variables:

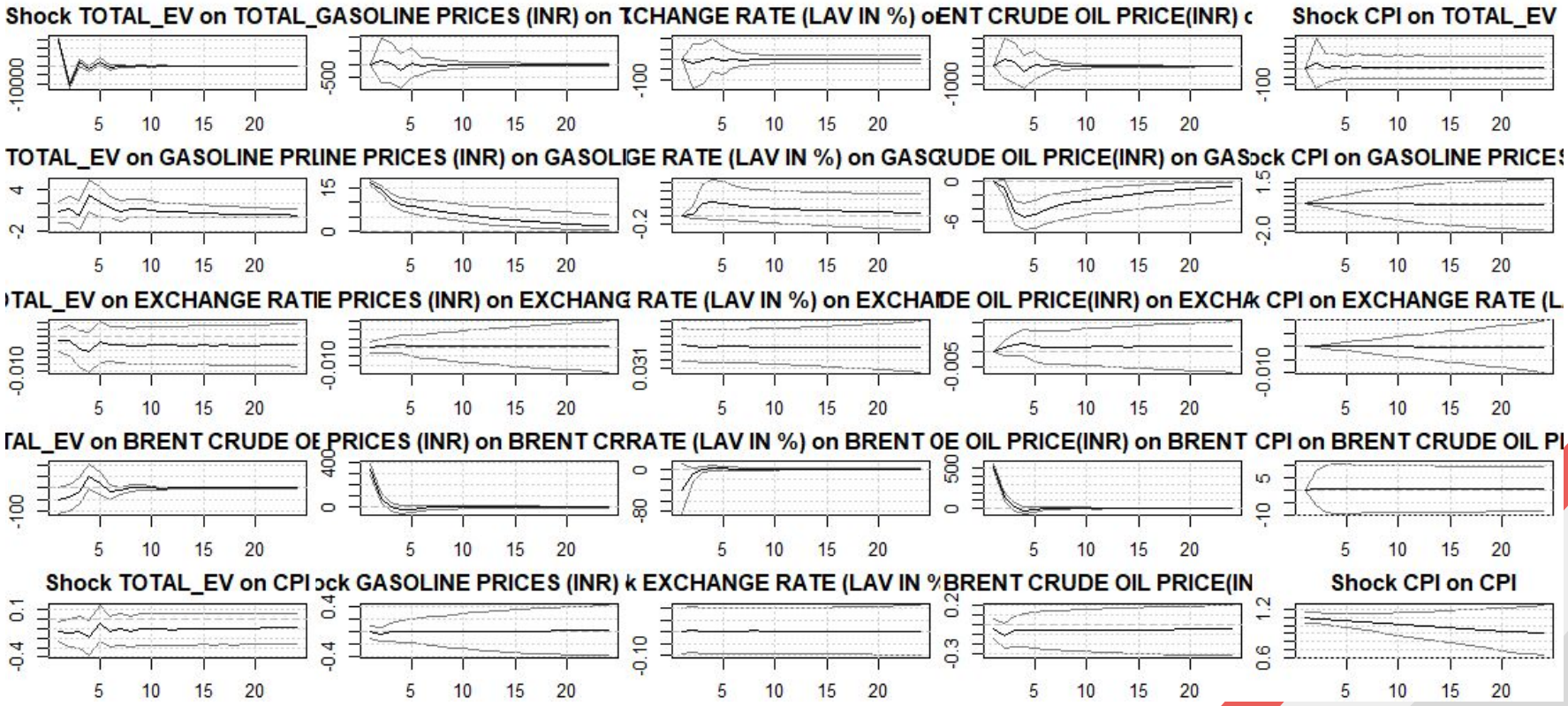
```
> print(dummy_effects)
             TOTAL_EV GASOLINE PRICES (INR) EXCHANGE RATE (LAV IN %)
FAME1       0.6970918          -0.0006374465              -1.812318e-06
FAME2       4.8978007          -0.0037921430               8.868100e-06
COVID       0.2372271          -0.0002512031              -2.201474e-06
RUS_UKR_WAR 0.3408273          -0.0011162594              -9.024258e-07
            BRENT CRUDE OIL PRICE(INR)           CPI
FAME1                     -0.02023929 -7.201771e-05
FAME2                      0.10104699 -2.032445e-04
COVID                     -0.01462926 -7.795452e-07
RUS_UKR_WAR               -0.06380327 -2.480987e-04
```

- Both FAME1 and FAME2 policies had significant positive impact of sales of EV, while COVID and Russia Ukraine war had small positive effect(model might capture after recovery phase of covid as well)

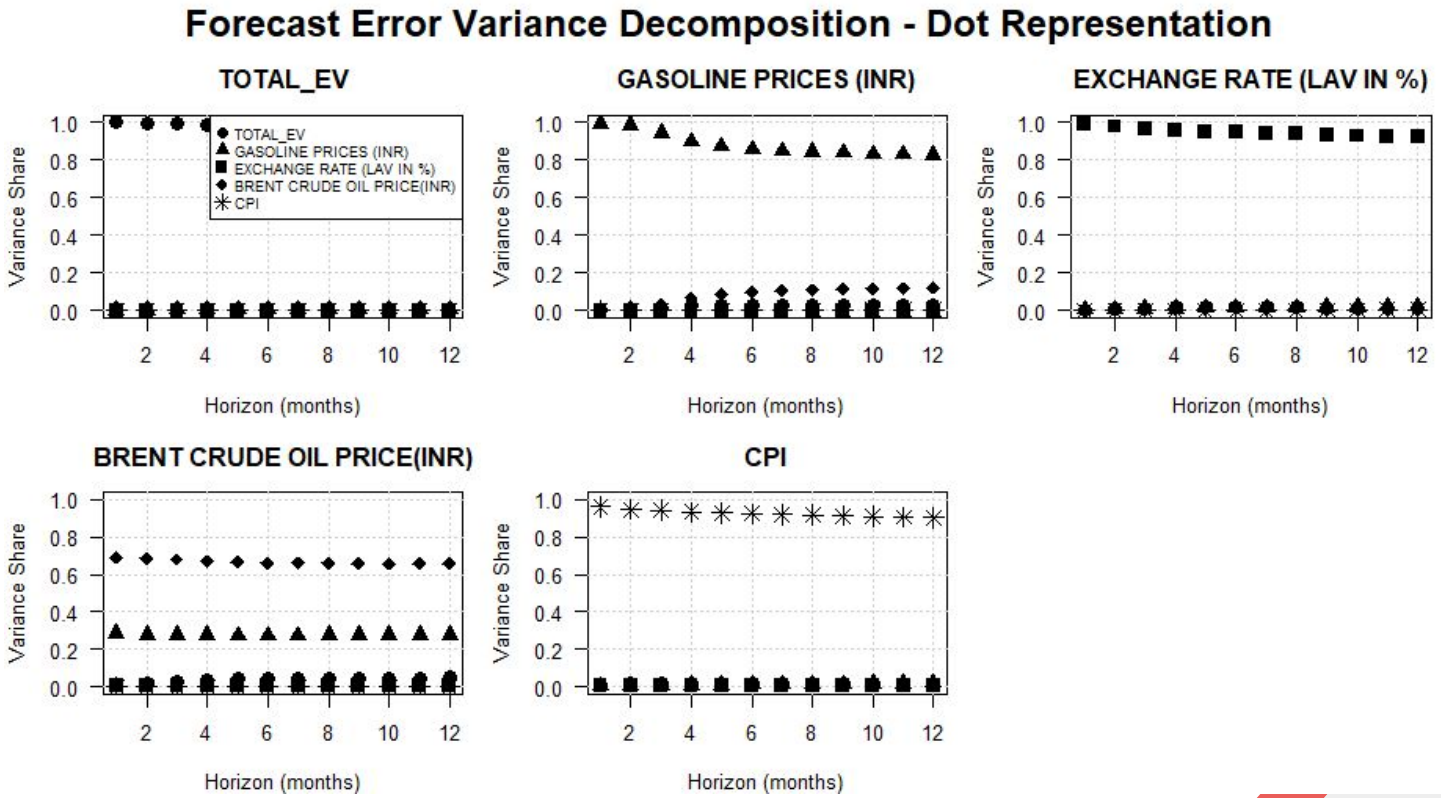
Granger Causality:

- Granger Causality test between GASOLINE PRICES→ TOTAL_EV sales show p- value = 0.04238 which is statistically significant indicating Gasoline Prices granger cause EV sales.
- Granger Causality test between TOTAL_EV → GASOLINE PRICES show p-value 0.4302 which is statistically insignificant indicating EV sales do not granger cause gasoline prices.
- This is consistent with the theory that when gasoline prices increase people may prefer to buy more electric vehicles. On the other hand, gasoline prices aren't affected much by EV sales.
- Therefore, a unidirectional relationship exist between gasoline prices and EV sales

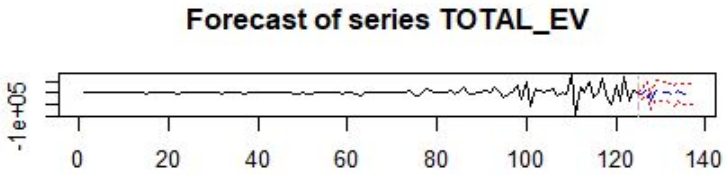
Impulse Response Function(IRF):

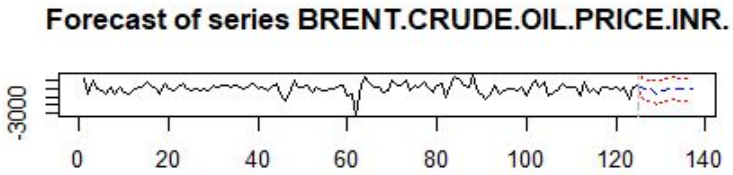


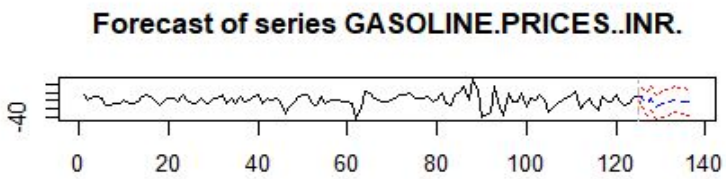
Forecast Error Variance Decomposition(FEVD):

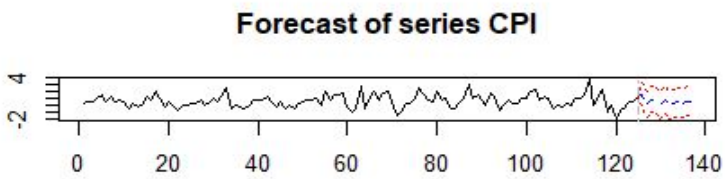


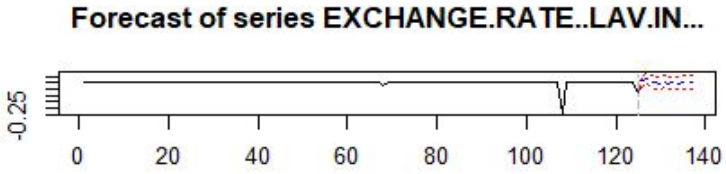
Forecasts:


Forecast of series TOTAL_EV
-1e+05
0
20
40
60
80
100
120
140


Forecast of series BRENT.CRUDE.OIL.PRICE.INR.
-3000
0
20
40
60
80
100
120
140


Forecast of series GASOLINE.PRICES..INR.
-40
0
20
40
60
80
100
120
140


Forecast of series CPI
-2
4
0
20
40
60
80
100
120
140


Forecast of series EXCHANGE.RATE..LAV.IN...
-0.25
0
20
40
60
80
100
120
140

Conclusion:

- All coefficients of EV sales in gasoline price equation are zero indicating that EV sales have no impact on gasoline prices in India
- Instead gasoline prices are affected by its own past value, lag of CPI indicating inflation effect and brent crude oil prices
- However, gasoline prices do impact sales of Electric Vehicles with delayed positive effect.
- The correlation between EV sales and Gasoline prices suggest that during the time of economic boost their is increase is sales of EV and higher demand for gasoline leading to increase in its price. This is a correlation and not a causation
- The results suggest that gasoline market is heavily influenced by global and macroeconomic factors like brent crude oil prices, exchange rate, cpi ,etc
- FAME1 and FAME2 adopted by Indian government to promote EV sales show significant positive impact on sales of EV, however they do not significantly affect gasoline prices
- The covid dummy shows slight positive effect on sales on EV, likely capturing post recovery phase effects
- The Russia Ukraine war dummy shows a positive effect on sales of EV likely due to the fact that war caused spike in global oil and gasoline prices making people shift towards EV. The slight negative effect on gasoline prices likely indicate structural break and main effects of war are likely captured by brent crude oil price.

Conclusion:

Research Question: Do the sales of EV (Electric Vehicles) affect the Gasoline Prices in India?
Answer: The sales of EV do not significantly affect gasoline prices in India (considering the small size of EV industry), but gasoline prices do affect EV sales.

References:

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- mospi.gov.in
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Thank You